

Performance Testing

Date	5 JULY 2026
Team ID	
Project Name	Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask Development Server (Browser Testing)
Type of Testing	Load Testing
Target Module / API	HDI Prediction Module (/predict) and Prediction History (/history)
Test Environment	Local Machine (macOS M4), Python 3.9, Flask, VS Code, Google Chrome
Test Date	5 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI for multiple countries	10	60	All predictions generated successfully
2	Access Prediction History page	20	60	History loads without delay
3	Submit multiple prediction requests simultaneously	30	90	System remains responsive
4	Navigate between Home, About, Result and History pages	15	60	Pages load successfully without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	≤ 1 second	0.62 second	Pass	Fast prediction
2	Response Time (Max)	≤ 3 seconds	1.45 seconds	Pass	Stable performance
3	Throughput (Req/ sec)	≥ 20 requests/ second	28 requests/ second	Pass	Handles concurrent users
4	Error Rate	0%	0%	Pass	No failed requests
5	CPU Utilization	≤ 50%	34%	Pass	Efficient CPU usage
6	Memory Utilization	≤ 60%	42%	Pass	Low memory consumption

Step 4: Observations & Analysis

The HDI Prediction System demonstrated stable performance during testing. Prediction requests were processed quickly with an average response time below one second. No request failures were observed during load testing. CPU and memory utilization remained well below the defined thresholds, indicating efficient resource usage. Navigation between project pages, including Home, Prediction, Result, History, and About, remained smooth even under concurrent user requests. Overall, the application successfully satisfied the performance requirements.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]



Human Development Index Predictor

Predict a country's Human Development Index (HDI) using Life Expectancy, Education and Gross National Income.

Country Name

Life Expectancy (Years)

Expected Years of Schooling

Mean Years of Schooling

GNI Per Capita (PPP)

Predict HDI

Prediction History

About Project

HDI Prediction Result

India

Indicator	Value
Life Expectancy	75.0 Years
Expected Years of Schooling	15.0 Years
Mean Years of Schooling	12.0 Years
GNI Per Capita (PPP)	\$50000.0

Predicted HDI Score

0.739

Development Category

High Human Development

[Predict Another Country](#)

[Prediction History](#)

[About Project](#)

Prediction History

Country	Life Expectancy	Expected Schooling	Mean Schooling	GNI Per Capita	HDI Score	Category
india	90.0	15.0	11.0	\$25000.0	0.628	Medium Human Development
pakistan	86.0	10.0	9.0	\$10000.0	0.564	Medium Human Development
india	70.0	10.0	11.0	\$200000.0	0.798	High Human Development
India	75.0	25.0	13.0	\$200000.0	0.847	Very High Human Development
india	75.0	15.0	12.0	\$50000.0	0.739	High Human Development

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